

Annexure 3b—Complete filing

INVENTION DISCLOSURE FORM

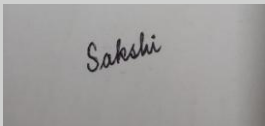
Details of Invention for better understanding:

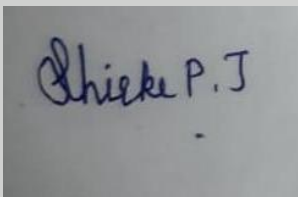
1. TITLE: Hybrid Solar-Wind and Piezo-Powered Smart Street Light

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2. INTERNAL INVENTOR(S)/STUDENT(S): All fields in this column are mandatory to be filled

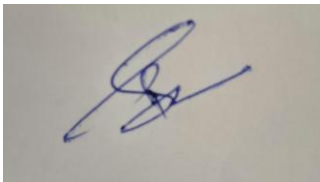
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For external inventors, a NOC (No Objection Certificate) from the affiliated institute/university/industry/lab, etc., is mandatory for each individual inventor and their respective topic. For NOC, the format is attached below.

(FOR ADDITIONAL INVENTORS, PLEASE ADD ROWS)

3. DESCRIPTION OF THE INVENTION: The prototype proposed the idea of a hybrid solar and wind renewable energy-powered smart street light system, which saves energy and runs without external power sources. It generates energy from three main power sources: from a solar photovoltaic generator, from a wind turbine generator, and from piezoelectrics to save energy. The invention improves the wastage of electrical energy and the disadvantages of multi-source power generation. During the daytime the system switches off automatically, saving electricity and manual power, and at nighttime it turns on only when the sensor detects motion.

The system consists of:

1. **Miniature wind turbine generator:** It produces power from wind, useful in hilly areas.
2. **Solar photovoltaic module:** It converts solar energy to electrical energy to charge the battery during the daytime.
3. **Piezoelectric cell:** It detects the pressure and movements and provides a high signal to the Arduino.
4. **TP4056 charging module:** a circuit for charging the Li-ion battery
5. **MT3608 boost converter:** to convert low power voltage from sensors and panel to stable and high power voltage for microcontroller and the lights.
6. **LDR (Light Dependent Resistor):** It determines whether day or night time, basically for the natural light level measure, sends the environment updates to the central control units.

7. **RCWL-0516 microwave Doppler radar sensor):** It detects the motion within the defined radius if there is movement it'll send updates hence the lights will turn on.

The system reduces energy consumption, improves sustainability, and is suitable for smart cities, remote areas, and energy-scarce environments.

- A. **PROBLEM ADDRESSED BY THE INVENTION:** Conventional street lighting systems mainly rely on electricity from the grid leading to costs and frequent outages during power failures. Most standalone solar or wind energy units rely on a power source causing them to be unreliable, in providing enough energy on cloudy days still wind conditions or extended use at night when stored power is insufficient. Additionally many systems do not incorporate control functions causing the lights to stay ON all night and squander energy when no movement is detected. In addition to that, most off-grid streetlights have poor battery protection, which reduces the life span of batteries and the overall system reliability. Urban and remote areas call for a more resilient, energy-efficient, and fully autonomous street lighting solution that is capable of operating continuously under varying environmental conditions and reducing useless energy consumption.

A. OBJECTIVE OF THE INVENTION (Provide minimum two)

1. Efforts should focus on creating an independent intelligent street lighting system driven by a hybrid mix of solar, wind, and piezoelectric energy sources to ensure uninterrupted functionality across various environmental scenarios.

2. Minimizing energy consumption through the use of intelligent lighting control by detecting ambient light and motion, which allows operation of the LEDs only when needed.

C. STATE OF THE ART/RESEARCH GAP/NOVELTY: How does your invention fulfill the research gap?

Sr. No.	Patent ID	Abstract	Research Gap	Novelty
1.	CN201546906U	Wind-solar hybrid generation system The system enables wind-sun complementary power generation, providing higher stability and reliability. The utility	No real-time smart streetlight features	No real-time smart streetlight features Our project has real-time implementation

		model presents a wind–solar hybrid generation system combining a wind generator and a photovoltaic panel on a single pole, both connected to a hybrid controller.		by powering streetlights.
2.	CN104333947A	The system comprises multiple single street lamp units, a road segment internet of things master control unit, and a monitoring center.	No provision for piezoelectric energy	Our invention has wind, solar, and piezoelectric energy implementation
3.	US8330296B2	Hybrid renewable energy turbine using wind and solar power An environmentally friendly combination of wind turbines and solar energy collectors is provided.	No IoT connectivity or remote monitoring	

D. DETAILED DESCRIPTION: The invention pertains to a renewable-energy intelligent street lighting system functioning independently by combining the invention. It involves a hybrid renewable-energy smart street lighting system that unites solar, wind and piezoelectric energy capture into one cohesive power unit. Energy gathered from a panel a wind turbine, and piezoelectric sensors is stored in a rechargeable lithium-ion battery managed through a TP4056 charging module and regulated by an MT3608 boost converter. The lighting unit consists of high-efficiency LEDs controlled by a MOSFET driver, while an LDR provides ambient-light detection and an RCWL-0516 microwave sensor enables motion-activated illumination. An Arduino microcontroller coordinates all sensing inputs, manages battery protection, and controls LED activation for intelligent operation, by which the system can provide continuous and energy-efficient lighting under different environmental conditions.

1. Hybrid Multi-Source Energy Harvesting System

This means that solar, wind, and piezoelectric energy generation methods are all incorporated into the same system so that solar panels will produce power from the sun and wind turbines can create electricity whenever the weather allows. The piezoelectric pad will convert the energy created from people walking over it and vehicles going over it into usable electrical energy. Using all three methods together creates an uninterrupted source of power.

2. Integrated Battery Storage and Power Management

The energy collected is stored inside the 18650 Lithium Ion Battery. The TP4056 Charging Module controls the charging process with overcharge protection and the ability to supply current to the battery. The MT3608 Boost Converter boosts battery voltage to the voltage level required to operate the Sensors, Microcontroller, and LED Headlights, allowing continuous operation regardless of varying input sources.

3. Smart Illumination Control Using Sensors

A light-dependent resistor or LDR is utilized by the light component of the project in order to measure and discern between the brightness of light packets (also known as “daylight”) and darkness packets (also known as “nighttime”). In this way, during nighttime hours the light will only be turned on when motion is detected from the microwave radar sensor (also called “microwave Doppler radar sensor”). This type of radar sensor operates in all weather conditions and allows the use of LEDs during times when they are actually needed. In turn, this provides a tremendous savings of energy.

4. Efficient LED Driver and Lighting Output

High-intensity LEDs are managed via a MOSFET switch serving as a low-loss driver. The MOSFET quickly toggles the LEDs on and off according to microcontroller instructions, reducing power loss and guaranteeing the LEDs provide peak brightness while using energy. This approach enhances the system efficiency and prolongs battery life.

5. Microcontroller-Based System Coordination and Protection

An Arduino microcontroller acts as the main control unit, receiving continuous input from the LDR, motion sensor, and battery voltage monitor. It executes programmed logic to decide when to activate lighting, when to conserve power, and how to prioritize energy from different sources.

The controller also prevents battery over-discharge, manages sensor timing, and optimizes total system performance under varying environmental conditions.

Block diagram and elaboration of components:

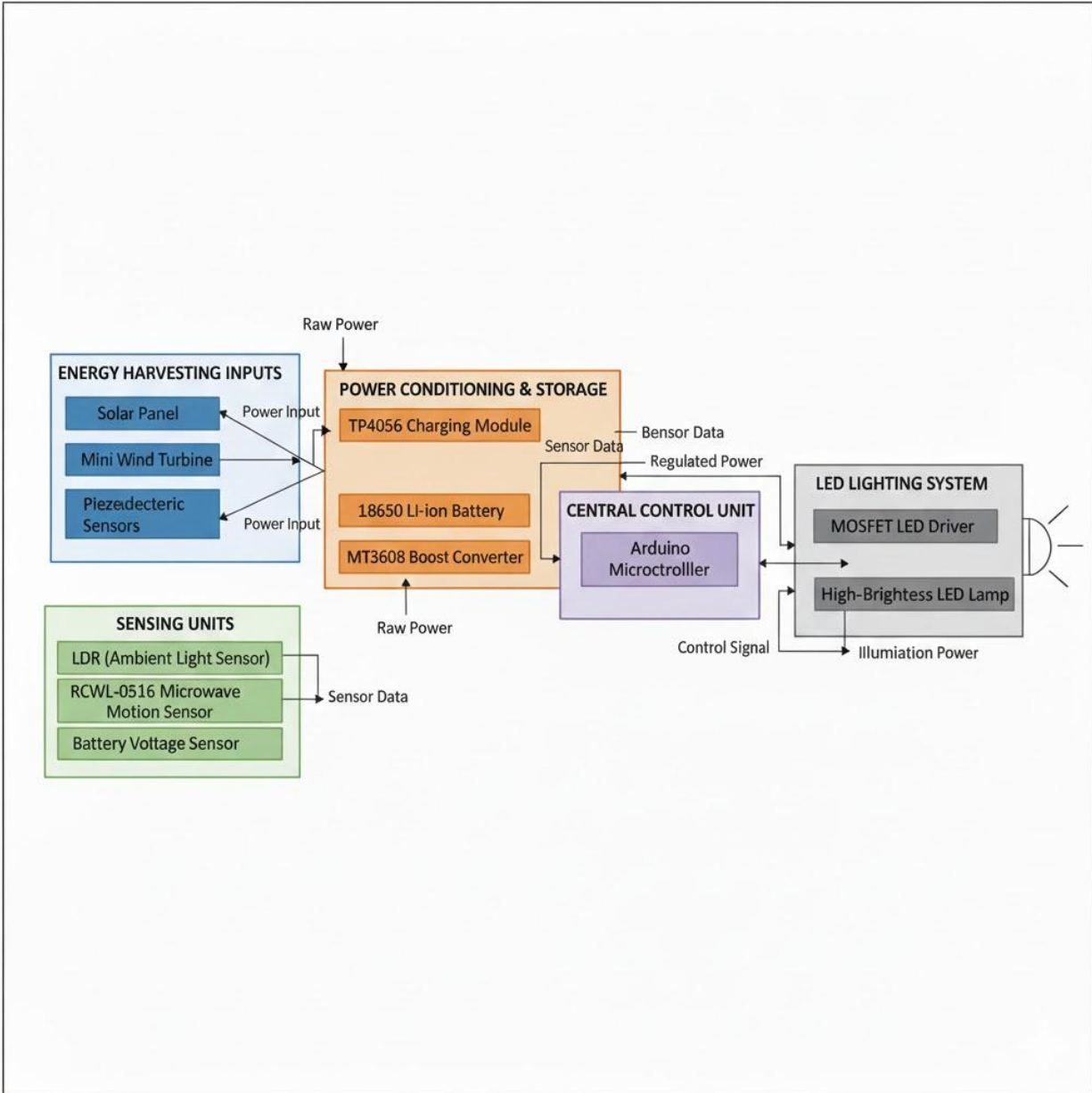


Fig. 1: Working of the prototype

Fig. 1 describes the prototype; it includes three energy harvesting components: a panel, a small wind turbine generator, and piezoelectric sensors that transform solar, wind, and mechanical energy into electrical energy. This generated power is then directed to the power conditioning and storage section, such as the TP4056 charging module, a circuit for charging the Li-ion

battery. Additionally there is a boost converter that elevates voltage energy to a higher and more stable voltage. Then there is the sensing unit, where we have three modules the first one is LDR module which detects and turns on the lights when sun sets and turns off automatically when sun rises, helping in less manual tasks. Then this all power inputs and sensor data goes through a microcontroller and process to turn on the street lights when it senses motion and also stores energy which makes it sustainable throughout.

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10. Overall Functional Operation

1. This invention comprises combined wind and piezoelectric elements that work together to start the operational process by harnessing energy at the same time. In daylight hours the primary energy source comes from the panel with the wind turbine and piezoelectric devices providing energy as required. Every type of harvested energy is directed to a lithium-ion battery, where the TP4056 charging module controls the charging to maintain energy storage. The obtained energy is increased to the designated operating voltage, by the MT3608 converter to supply power to the sensing units, control electronics and LED lighting system.
2. Throughout the daytime the entire system remains in a low-power monitoring state because the LDR detects surrounding light and prevents the lighting from turning ON. At night when the ambient light falls beneath the threshold the microcontroller transitions to night mode. In this state the RCWL-0516 microwave motion sensor constantly observes the area, for any vehicle activity. Upon sensing motion the sensor transmits a trigger command to the microcontroller, which subsequently powers the light linked via the MOSFET for a certain duration.
3. If the system does not detect any movement, it keeps the LEDs off or in a standby mode, thus saving energy. During all operation processes, the microcontroller continuously checks the battery voltage to prevent deep discharging and controls lighting activity depending on available power. Integration of multi-source energy harvesting with advanced sensing and decision-making by a microcontroller will ensure consistent and efficient autonomous street lighting in a wide range of environmental conditions.

E. RESULTS AND ADVANTAGES:

This is a Smart Streetlight Hybrid Wind and Piezoelectric System, which is designed to operate continuously and reliably in any weather condition through the use of multiple renewable energy sources. From test results on the amount of

energy collected from the panels in conjunction with the output of the turbines and the generation of power through piezoelectricity, the system has enough stored energy in its battery to maintain a steady charge throughout the 24-hour cycle.

The motion-detection lighting controls greatly cut down on the amount of energy consumed, as the LED's are only powered on when there is either a human or vehicle detected. The LDRs used to perform the day-night transition for lighting accurately sense ambient light, and the microcontroller is responsible for controlling power usage, keeping the batteries safe, and coordinating the use of all sensors.

Outlined below are a number of benefits offered by the invention:

1. **Continuous Operation without Grid Link:** Combining three power sources ensures the system functions during weather periods of low wind or when foot traffic is scarce thus improving reliability compared to systems dependent, on only one energy source.
2. **Notable Energy Savings:** By preventing the lights from staying on during the night motion-sensor lighting reduces power usage greatly enhances battery lifespan and decreases energy demand..
3. **Improved Detection Ability:** Utilizing the microwave Doppler radar sensor increases the accuracy and range of motion detection, which may fall short when using sensors commonly found in street lighting setups.
4. **Enhanced Battery. Safety:** The TP4056 charging unit combined with microcontroller-driven power monitoring protects the battery from overcharging, excessive discharging, and voltage fluctuations, thereby extending its lifespan.
5. **Scalability and Suitability for Smart Cities:** The invention is perfect for smart cities, remote areas, and places with no electrical infrastructure due to its autonomous and maintenance-efficient design.

6. **Novel Energy Supplementation:** The integration of piezoelectric harvesting adds a supplementary micro-energy input that enhances system resilience, lending novelty and what could be considered a patentable aspect to the invention.

F. EXPANSION:

1. Maximum Power Point (MPPT) tracking enables greater extraction of energy from both wind turbines by providing highly-developed MPPT controllers that enable total power to be maximised during quickly-changing weather patterns (i.e. unstable weather).
2. Advanced Piezoelectric Systems could be installed to take advantage of all of the mechanical energy generated in heavy pedestrian areas (like urban sidewalks) so that more supplemental energy can be supplied through the addition of more units (Greater energy generation = more units).
3. Future versions of the device may contain wireless technologies such as LoRa, WiFi, and NB-IoT, which will allow users to monitor, log data, detect failures, and operate these smart-city devices through their respective networks.
4. Machine learning-based Adaptive Lighting Algorithms enable the automatic adjustment of the brightness of lighting fixtures depending on the number of pedestrians present, the amount of battery power available, and the current ambient conditions.

Integration with Smart Grids: The device will return the energy harvested from the sun to either the microgrid or to a shared storage facility, thereby improving the energy efficiency of all types of lighting throughout the facility. Will self-clean; The device will be built with a strong housing and hydrophobic coatings for water and to allow the device to withstand the elements and the effects of increased vibration from wind/ice, etc., thus ensuring operation in an all-weather outdoor environment. Modular and scalable; The Modular components can be built to support additional factors (renewable energy sources (solar, wind, etc.), additional battery storage capacity, advanced sensor capabilities and additional components) required by the application. Hybrid sensor fusion; Many different types of sensor technology (passive infrared (PIR), ultrasonic, infrared thermal imaging and visual display) can be used in complex environments to assist in the installation of lighting equipment and eliminate false triggers.

G. WORKING PROTOTYPE/FORMULATION/DESIGN/COMPOSITION:

Fig. 2 illustrates the prototype featuring three energy harvesting elements: a panel, a wind turbine generator, and piezoelectric sensors that convert solar, wind, and mechanical energy into electrical power. This produced electricity is subsequently routed to the power conditioning and storage unit, like the TP4056 charging module, a circuit designed to charge the Li-ion battery. Moreover, a boost converter is included to increase the voltage to a more stable level. Then there is the sensing unit, where we have three modules. The first one is the LDR module, which detects and turns on the lights when the sun sets and turns off automatically when the sun rises, helping in less manual tasks. Then all these power inputs and sensor data go through a microcontroller and process to turn on the streetlights when it senses motion and also store energy, which makes it sustainable throughout.

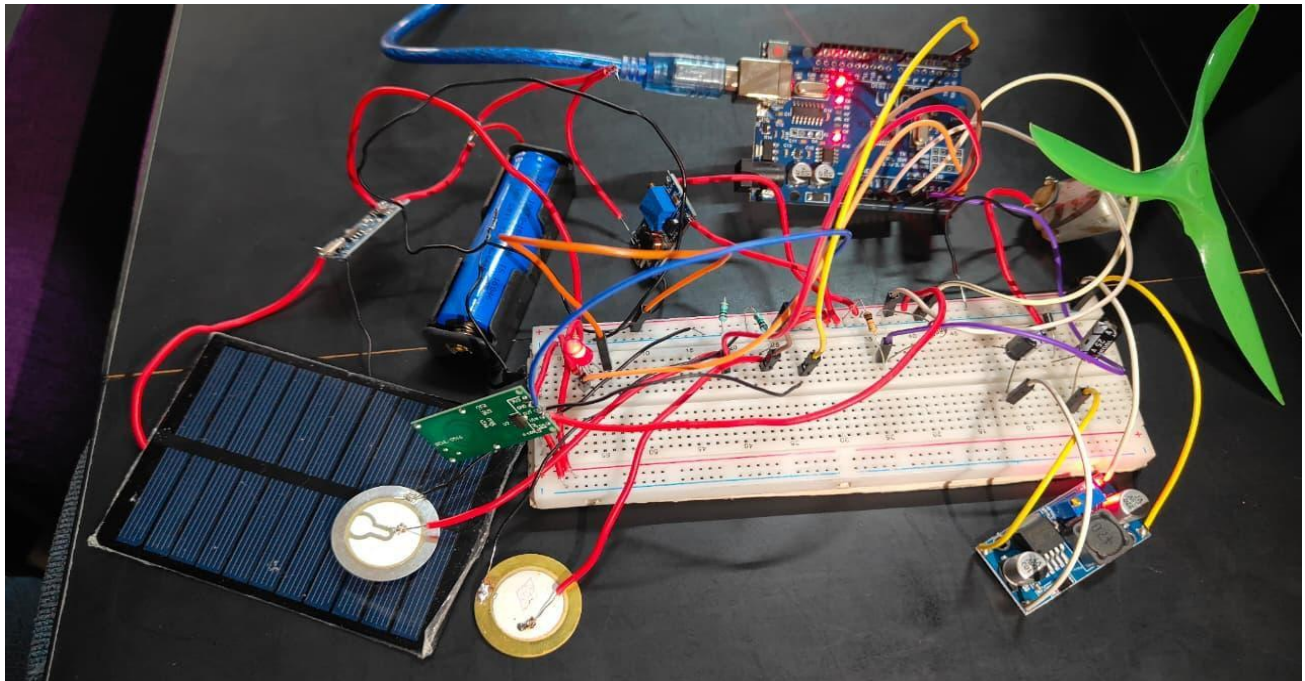


FIG. 2: HARDWARE PROTOTYPE

7. Overall Comparative Summary

1. **Source hybrid energy system:** Unlike lighting that relies solely on solar power or the electrical grid this innovation combines solar, wind and piezoelectric energy collection significantly improving energy reliability.
2. **Operation in adverse weather:** Traditional solar streetlights cease functioning during rainy conditions but the addition of wind and piezoelectric energy sources allows continuous operation even, with limited sunlight.

3. **Superior motion detection technology:** Current systems primarily rely on PIR sensors that have an outdoor detection range; this invention utilizes a microwave Doppler sensor to achieve greater precision and improved adaptability to environmental conditions.
4. **Smart lighting management:** Traditional systems typically keep lights on all night while this gadget activates LEDs when motion is detected resulting in significant power savings.
5. **Improved battery control:** Many low-cost models lack battery protection; this invention uses TP4056 regulation and microcontroller monitoring to prevent overcharging and deep discharge.
6. **Higher system efficiency:** The MOSFET-based LED driver minimizes switching losses, providing more efficient illumination than resistor- or relay-based LED driving methods.
7. **Enhanced operational independence:** Unlike street lamps on external electricity or a single power source this innovation functions fully, off-grid and is entirely autonomous.
8. **Incorporation of energy harvesting:** Conventional designs overlook energy capture; this innovation integrates piezo units to gather energy from pedestrian or vehicular movement enhancing longevity.
9. **Environmental responsiveness:** Incorporating an LDR to identify day and night in conjunction, with motion sensing allows the system to respond dynamically unlike lighting setups that rely on fixed schedules or stay on continuously.
10. **Suitability for smart city deployment:** Compared to older manual or semi-automatic systems, the invention offers modernized, low-maintenance, autonomous functionality ideal for smart infrastructure and remote locations.

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/design to anyone or in any conference?	YES ()	NO (*)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO (*)
C. Has your invention been described in any printed publication or any other form of media, such as the Internet?	YES ()	NO (*)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO (*)
E. Name of regulatory body or any other approvals if required.	YES ()	NO (*)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. **NA**

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside the university (any industry, other universities, or any other entity). **NA**

7. Potential Chances of Commercialization. **YES**

8. List of companies that can be contacted for commercialization along with the website link.

Company	Specialization	Website
Aura energy	Hybrid solar grid energy	https://www.auraenergy.com/hybridsolar-street-lights
JioThings (Jio)	Offers smart street lighting solutions for energy optimization and fault detection in India.	https://www.jio.com/business/services/iot/smart-street-lighting/

Company	Specialization	Website
Inlux Solar	Wide range of hybrid solar-wind streetlights.	https://www.inluxsolar.com/product/solar-wind-hybrid-street-light/

9. Any basic patent that has been used, and we need to pay a royalty to them. **NA**

10. FILING OPTIONS: Based on the current stage of development, where the design, technical structure, and functional concept of the BLDC Fast Hand Fan are fully established and a prototype can be developed within a short duration, the invention is eligible for Provisional Patent Filing.

- **Complete Specification Filing: ✓ Recommended**
Can be submitted after prototype testing, performance validation, and final technical drawings are completed.

11. KEYWORDS:

Hybrid energy harvesting, solar-powered street light, wind turbine generator, piezoelectric energy harvesting, smart lighting system, motion-activated illumination, microwave Doppler sensor, RCWL-0516, ambient light detection, LDR sensor, microcontroller-based control, Arduino, Li-ion battery, TP4056 charger, MT3608 boost converter, MOSFET LED driver, renewable energy system, off-grid lighting, energy-efficient street light, smart city infrastructure.

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

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